Chapter 3: Discrete Random Variables

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CSAI2121 Probability and Statistics

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Learning Objectives

By the end of this lecture, you should be able to:

- Define a random variable and distinguish discrete vs. continuous
- Construct and interpret a probability mass function (PMF)
- Construct and interpret a cumulative distribution function (CDF)
- Compute the expected value and variance of a discrete random variable
- Apply linearity of expectation
- Recognize and apply the Bernoulli, Binomial, and Poisson distributions

Random Variables

Motivation: Why Random Variables?

- Last week: events were named outcomes ("disease", "spam").
- Today: we assign **numbers** to outcomes, enabling arithmetic — averages, sums, comparisons.
- Example: "How many typos does a generated sentence contain?" is more useful for analysis than "Does it contain a typo?"
- This numerical view is essential for expectation, variance, and eventually MLE (Week 8).

Definition of a Random Variable

Definition

A **random variable** X is a function that assigns a real number to each outcome in the sample space S:

$$X : S \to \mathbb{R}$$

- Notation: capital letters (X, Y) denote the random variable; lowercase (x, y) denote specific values it can take.
- $P(X = x)$ reads "the probability that X takes the value x".

Discrete vs. Continuous (Preview)

- **Discrete**: countable set of possible values (e.g., number of heads, number of errors).
- **Continuous**: uncountably infinite values on an interval (e.g., response time, temperature) — covered next week.

This week

We focus entirely on **discrete** random variables.

Worked Example 1: Coin Flips as a Random Variable

Flip a fair coin 3 times. Let X = number of heads.

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Outcome	X (number of heads)
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HHT, HTH, THH	2
HTT, THT, TTH	1
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Range of X: $\{0, 1, 2, 3\}$

Worked Example 2: Random Variable in a Dataset (AI Context)

Let X = number of misclassified images in a batch of 5 test images.

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Let X = number of misclassified images in a batch of 5 test images.

- Possible values: $\{0, 1, 2, 3, 4, 5\}$
- X converts a complex outcome (which specific images failed) into a single useful number.
- This is exactly how we will later compute an "error count" metric for a classifier.

Probability Mass Function

Definition of PMF

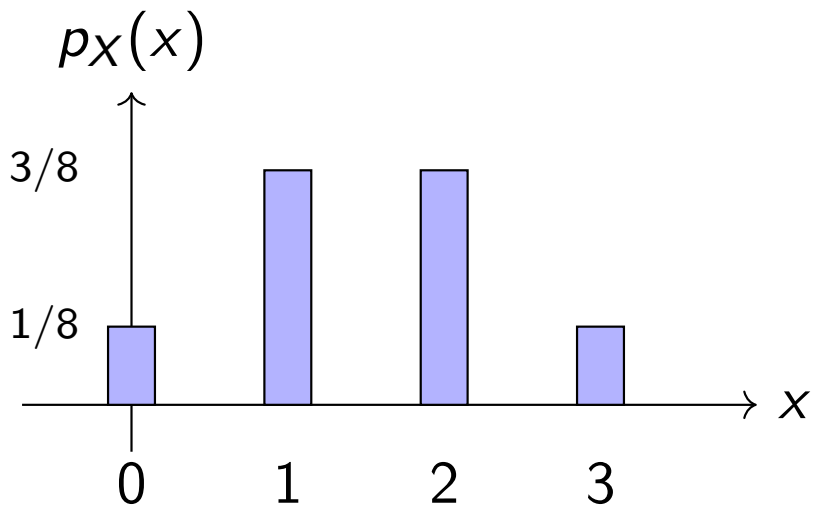
Definition

The **probability mass function** (PMF) of a discrete random variable X is

$$p_X(x) = P(X = x)$$

- Property 1: $p_X(x) \geq 0$ for all x
- Property 2: $\sum_x p_X(x) = 1$

Visual Intuition: PMF as a Bar Chart



- Each spike represents the probability of one specific value.
- Contrast with the continuous case next week: bars vs. a smooth curve.

Worked Example 3: PMF of Number of Heads (3 Flips)

x	0	1	2	3
$p_X(x)$	1/8	3/8	3/8	1/8

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$$\sum_x p_X(x) = \frac{1}{8} + \frac{3}{8} + \frac{3}{8} + \frac{1}{8} = 1 \quad \checkmark$$

This is exactly the bar chart shown on the previous slide.

Worked Example 4: PMF of Model Errors (AI Context)

A classifier is applied to 3 independent test cases, each with $P(\text{error}) = 0.2$. Let $X =$ number of errors out of 3.

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x	0	1	2	3
$p_X(x)$	0.512	0.384	0.096	0.008

Note

This is actually a **Binomial** distribution — we will formalize this later today.

Class Exercise (Pair Work)

Problem

A fair die is rolled. Let $X = 1$ if the roll is prime $(2, 3, 5)$, and $X = 0$ otherwise.

1. Construct the PMF of X.
2. Verify that the PMF sums to 1.

Work in pairs (3 minutes), then we discuss as a class.

Instructor Demo 1: PMF of a Loaded Die

A loaded die has $P(6) = 0.5$ and each of 1–5 has probability 0.1.

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x	1	2	3	4	5	6
$p_X(x)$	0.1	0.1	0.1	0.1	0.1	0.5

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x	1	2	3	4	5	6
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$$\sum_x p_X(x) = 5(0.1) + 0.5 = 1 \quad \checkmark$$

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$p_X(x)$	0.1	0.1	0.1	0.1	0.1	0.5

$$\sum_x p_X(x) = 5(0.1) + 0.5 = 1 \quad \checkmark$$

This is exactly the "loaded die" from Chapter 2's Bayes' Theorem demo — same object, now described using a PMF.

Cumulative Distribution Function

Definition of CDF

Definition

The **cumulative distribution function** (CDF) of X is

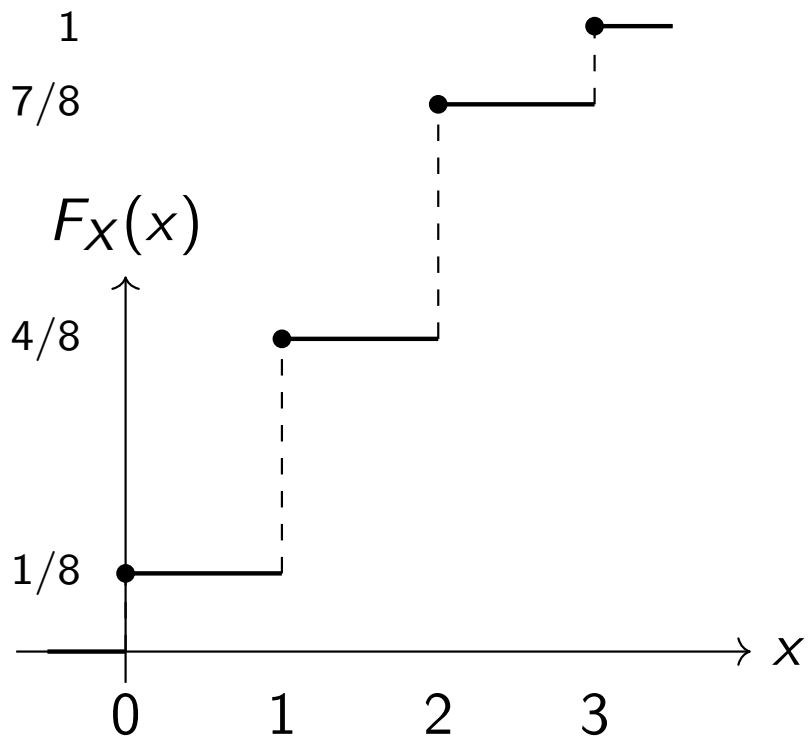
$$F_X(x) = P(X \leq x)$$

For discrete X:

$$F_X(x) = \sum_{k \leq x} p_X(k)$$

- F_X is non-decreasing
- $F_X(-\infty) = 0, \quad F_X(\infty) = 1$

Visual Intuition: CDF as a Step Function



Jumps occur exactly at values with positive PMF; jump size = PMF value at that point.

Worked Example 5: CDF from the Coin-Flip PMF

Recall the 3-coin-flip PMF: $p_X(0) = 1/8$, $p_X(1) = 3/8$, $p_X(2) = 3/8$, $p_X(3) = 1/8$.

Worked Example 5: CDF from the Coin-Flip PMF

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x	0	1	2	3
$F_X(x)$	$1/8$	$4/8$	$7/8$	1

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Recall the 3-coin-flip PMF: $p_X(0) = 1/8$, $p_X(1) = 3/8$, $p_X(2) = 3/8$, $p_X(3) = 1/8$.

x	0	1	2	3
$F_X(x)$	$1/8$	$4/8$	$7/8$	1

$$P(X \leq 2) = F_X(2) = \frac{7}{8}, \qquad P(X \geq 2) = 1 - F_X(1) = 1 - \frac{4}{8} = \frac{4}{8}$$

Recovering the PMF from the CDF

Fact

$$p_X(x) = F_X(x) - F_X(x^-)$$

(the size of the jump at x)

- Useful when only the CDF is given.
- Example: from the table above, $p_X(2) = F_X(2) - F_X(1) = \frac{7}{8} - \frac{4}{8} = \frac{3}{8}$ — matches the original PMF.

Class Exercise

Problem

Using the PMF from the earlier exercise ($X = 1$ if a die roll is prime, else $X = 0$):

1. Construct the CDF $F_X(x)$.
2. Compute $P(X \leq 0.5)$.
3. Compute $P(X \leq 1)$.

Expected Value

Motivation: The "Center of Mass"

- If we repeated the experiment many times and averaged the outcomes, what number would we converge to?
- This idea connects directly to the **Law of Large Numbers** (Week 7).

Definition of Expected Value

Definition

$$E[X] = \sum_x x \cdot p_X(x)$$

- Interpretation: a weighted average, with weights given by the probabilities.
- Also called the **mean** of X, often denoted μ.

Worked Example 6: Expected Value of a Die Roll

$$E[X] = \sum_{x=1}^{6} x \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Worked Example 6: Expected Value of a Die Roll

$$E[X] = \sum_{x=1}^{6} x \cdot \frac{1}{6} = \frac{1+2+3+4+5+6}{6} = \frac{21}{6} = 3.5$$

Key Point

3.5 is **not** a possible outcome of a single roll — expectation is a long-run average, not a "typical" single value.

Worked Example 7: Expected Number of Clicks (AI Context)

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$$E[X] = \sum_{x=0}^{3} x \cdot p_X(x) = 0.24$$

On average, the user is expected to click 0.24 times across 3 impressions.

Expectation of a Function of X

Definition

$$E[g(X)] = \sum_x g(x) \cdot p_X(x)$$

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Mini-example: for the die roll,

$$E[X^2] = \sum_{x=1}^{6} x^2 \cdot \frac{1}{6} = \frac{1+4+9+16+25+36}{6} = \frac{91}{6} \approx 15.17$$

We will need $E[X^2]$ shortly, to compute variance.

Linearity of Expectation

Key Property

$$E[aX + b] = aE[X] + b$$

$$E[X + Y] = E[X] + E[Y] \quad \text{(even if } X, Y \text{ are dependent!)}$$

- One of the most powerful tools in probability.
- The sum rule requires **no independence assumption** at all.

Worked Example 8: Combining Two Model Costs (AI Context)

Model A makes X errors per batch with $E[X] = 2$; Model B makes Y errors with $E[Y] = 3$.

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Important

This holds even if X and Y are correlated (e.g., both models tend to fail on the same hard examples).

Variance and Standard Deviation

Motivation: Measuring Spread

- Two distributions can have the same mean but very different spread.
- Example: two classifiers with the same average accuracy but very different consistency across batches.
- Expectation alone does not capture this — we need **variance**.

Definition of Variance

Definition

$$\text{Var}(X) = E\left[(X - E[X])^2\right] = E[X^2] - \left(E[X]\right)^2$$

- Standard deviation: $\text{SD}(X) = \sqrt{\text{Var}(X)}$
- Same units as X itself (unlike variance).

Worked Example 9: Variance of a Die Roll

From before: $E[X] = 3.5$, $E[X^2] = \frac{91}{6} \approx 15.17$

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From before: $E[X] = 3.5$, $E[X^2] = \frac{91}{6} \approx 15.17$

$$\text{Var}(X) = 15.17 - (3.5)^2 = 15.17 - 12.25 \approx 2.92$$

$$\text{SD}(X) = \sqrt{2.92} \approx 1.71$$

Properties of Variance

Key Property

$$\text{Var}(aX + b) = a^2 \,\text{Var}(X)$$

Common Exam Trap

Shifting ($+b$) does **not** change spread. Scaling ($\times a$) changes spread by a factor of a^2 — **not** a. Compare with $E[aX + b] = aE[X] + b$, which is linear in a.

Class Exercise

Problem

Given the PMF: $P(X = 0) = 0.5$, $P(X = 1) = 0.3$, $P(X = 2) = 0.2$.

1. Compute $E[X]$.
2. Compute $E[X^2]$.
3. Compute $\text{Var}(X)$ and $\text{SD}(X)$.

Work individually (5 minutes).

The Bernoulli Distribution

Motivation: Binary Outcomes

- Countless AI/ML scenarios reduce to yes/no: spam/not spam, click/no-click, correct/incorrect prediction.
- We now formalize this as the simplest possible discrete distribution.

Definition: Bernoulli Distribution

Definition

$X \sim \text{Bernoulli}(p)$: $\quad P(X = 1) = p, \quad P(X = 0) = 1 - p$

$$p_X(x) = p^x(1-p)^{1-x}, \qquad x \in \{0, 1\}$$

Mean and Variance of Bernoulli

$$E[X] = 0 \cdot (1-p) + 1 \cdot p = p$$

$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

Mean and Variance of Bernoulli

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$$\text{Var}(X) = E[X^2] - (E[X])^2 = p - p^2 = p(1-p)$$

Interesting Fact

Variance is maximized at $p = 0.5$ (maximum uncertainty about the outcome) — foreshadows the entropy discussion in Week 10.

Worked Example 10: Binary Classifier Output (AI Context)

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$$E[X] = 0.9, \qquad \text{Var}(X) = (0.9)(0.1) = 0.09$$

Worked Example 10: Binary Classifier Output (AI Context)

A spam filter flags an email correctly with probability $p = 0.9$. Let $X = 1$ if correct.

$$E[X] = 0.9, \qquad \text{Var}(X) = (0.9)(0.1) = 0.09$$

Interpretation: even a "good" classifier has non-trivial variance for a single prediction.

Instructor Demo 2: Bernoulli in an A/B Test

A website tests a new button design. Each visitor converts (makes a purchase) with $P(\text{convert}) = 0.05$. Let $X = 1$ if a randomly chosen visitor converts.

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$$E[X] = 0.05, \qquad \text{Var}(X) = (0.05)(0.95) = 0.0475$$

This single Bernoulli variable is the building block for the Binomial distribution we meet after the break — summing many such independent visitors.

The Binomial Distribution

Motivation: Repeated Independent Trials

- What if we repeat a Bernoulli trial n times and count the number of successes?
- Requirements:
  - Fixed number of trials n
  - Trials are independent
  - Constant success probability p across trials

Definition: Binomial Distribution

Definition

$X \sim \text{Binomial}(n, p)$:

$$P(X = k) = \binom{n}{k} p^k (1-p)^{n-k}, \qquad k = 0, 1, \ldots, n$$

- $\binom{n}{k}$ counts the number of arrangements with k successes
- $p^k(1-p)^{n-k}$ is the probability of *one specific* arrangement

Mean and Variance of Binomial

Key Result

$$E[X] = np, \qquad \text{Var}(X) = np(1-p)$$

Mean and Variance of Binomial

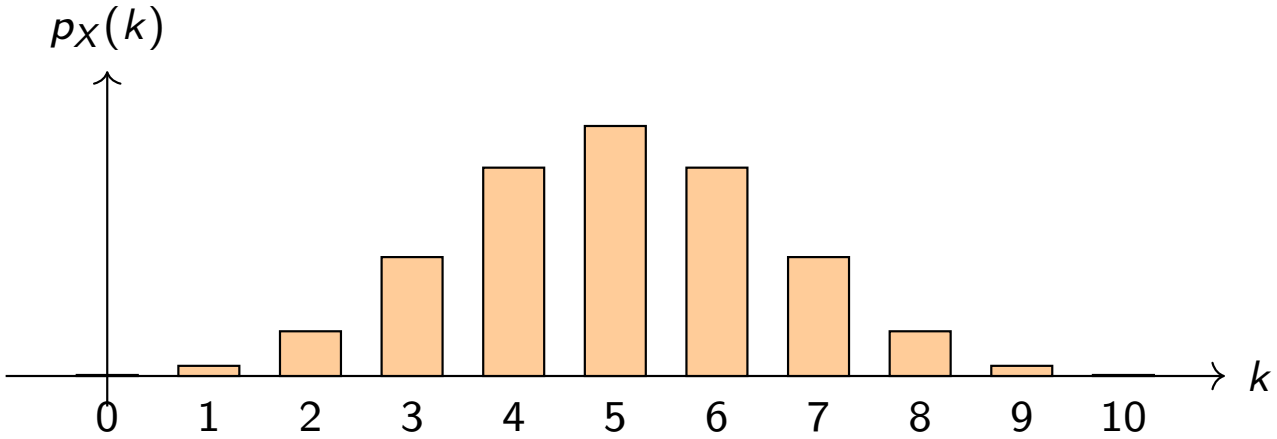
Key Result

$$E[X] = np, \qquad \text{Var}(X) = np(1-p)$$

Quick justification via linearity: write $X = X_1 + X_2 + \cdots + X_n$, where each $X_i \sim \text{Bernoulli}(p)$ independently.

$$E[X] = \sum_{i=1}^{n} E[X_i] = np \qquad \text{(Slide 23's linearity rule)}$$

Visual: Binomial PMF Shape ($n = 10,\ p = 0.5$)



- Symmetric when $p = 0.5$; skewed when $p \neq 0.5$ (e.g., left-skewed toward small k if $p = 0.2$, right-skewed if $p = 0.8$).

Worked Example 11: Quality Control

A batch of 10 components, each defective independently with $p = 0.05$.
Let $X =$ number defective.

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$$E[X] = 10(0.05) = 0.5, \qquad \text{Var}(X) = 10(0.05)(0.95) = 0.475$$

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A batch of 10 components, each defective independently with $p = 0.05$. Let $X =$ number defective.

$$E[X] = 10(0.05) = 0.5, \qquad \text{Var}(X) = 10(0.05)(0.95) = 0.475$$

$$P(X = 0) = \binom{10}{0}(0.05)^0(0.95)^{10} \approx 0.599$$

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$$P(X = 0) = \binom{10}{0}(0.05)^0(0.95)^{10} \approx 0.599$$

$$P(X \leq 1) = P(X = 0) + P(X = 1) \approx 0.599 + 0.315 = 0.914$$

Worked Example 12: Correct Predictions Out of n (AI Context)

A model has per-query accuracy $p = 0.85$; 20 independent queries. Let X = number correct.

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$$E[X] = 20(0.85) = 17, \qquad \text{Var}(X) = 20(0.85)(0.15) = 2.55$$

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$$E[X] = 20(0.85) = 17, \qquad \text{Var}(X) = 20(0.85)(0.15) = 2.55$$

Question: $P(X \geq 18)$?

$$P(X \geq 18) = P(X = 18) + P(X = 19) + P(X = 20)$$

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Question: $P(X \geq 18)$?

$$P(X \geq 18) = P(X = 18) + P(X = 19) + P(X = 20)$$

Each term is computed with the Binomial PMF formula — summing a few terms is often easier than it looks.

Class Exercise

Problem

A recommendation system shows an ad to 8 users; each clicks independently with $p = 0.25$. Let $X =$ number of clicks.

1. Compute $P(X = 2)$.
2. Compute $E[X]$.
3. Compute $\text{Var}(X)$.

Instructor Demo 3: Binomial and the Multiplication Rule

Recall Chapter 2, Instructor Demo 2: a component passes Stage 1 with $P = 0.90$, and Stage 2 (given Stage 1) with $P = 0.80$, giving $P(\text{pass both}) = 0.72$.

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$$X \sim \text{Binomial}(5,\ 0.72)$$

$$E[X] = 5(0.72) = 3.6, \qquad \text{Var}(X) = 5(0.72)(0.28) \approx 1.01$$

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$$X \sim \text{Binomial}(5,\ 0.72)$$

$$E[X] = 5(0.72) = 3.6, \qquad \text{Var}(X) = 5(0.72)(0.28) \approx 1.01$$

This shows how Chapter 2's multiplication rule directly feeds into this week's Binomial parameter p.

The Poisson Distribution

Motivation: Counting Rare Events

- What if n is very large and p is very small, but np stays moderate?
- Examples:
  - Number of server requests per minute
  - Number of typos per page
  - Number of API failures per hour

Definition: Poisson Distribution

Definition

$X \sim \text{Poisson}(\lambda)$:

$$P(X = k) = \frac{\lambda^k e^{-\lambda}}{k!}, \qquad k = 0, 1, 2, \ldots$$

- $\lambda =$ average rate of occurrence over the interval of interest.

Mean and Variance of Poisson

Key Result

$$E[X] = \lambda, \qquad \text{Var}(X) = \lambda$$

Distinctive Property

The mean equals the variance — a hallmark of the Poisson distribution, useful for quickly checking if data are "Poisson-like".

Poisson as a Limit of Binomial

- As $n \to \infty$ and $p \to 0$, with $np = \lambda$ held fixed:

$$\text{Binomial}(n, p) \longrightarrow \text{Poisson}(\lambda)$$

- Practical rule of thumb: use Poisson when n is large and p is small (e.g., $n > 50$, $p < 0.05$).

No derivation required — this is a statement you should recognize and use.

Worked Example 13: Server Requests

A server receives requests at an average rate $\lambda = 4$ per minute.

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$$P(X = 2) = \frac{4^2 e^{-4}}{2!} = \frac{16 e^{-4}}{2} \approx 0.1465$$

$$P(X \geq 1) = 1 - P(X = 0) \approx 1 - 0.0183 = 0.9817$$

Worked Example 14: API Error Monitoring (AI Context)

An ML inference API fails on average $\lambda = 1.5$ times per hour.

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This models rare, independent failure events — exactly the kind of quantity used in production ML monitoring and alerting systems.

Class Exercise

Problem

A content-moderation system flags an average of 3 posts per hour as violations.

1. Compute $P(X = 5)$.
2. Compute $P(X = 0)$.

Instructor Demo 4: Comparing Binomial and Poisson Approximation

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Observation

The two models agree almost perfectly — exactly the regime (*large* n, *small* p) where the Poisson approximation is justified.

Summary and Looking Ahead

Summary Table: Three Key Distributions

Distribution	PMF	Mean	Variance	Typical Use
Bernoulli(p)	$p^x(1-p)^{1-x}$	p	$p(1-p)$	single yes/no trial
Binomial(n, p)	$\binom{n}{k}p^k(1-p)^{n-k}$	np	$np(1-p)$	# successes in n trials
Poisson(λ)	$\lambda^k e^{-\lambda}/k!$	λ	λ	# rare events in an interval

Common Mistakes

- Confusing $E[X]$ with the *most likely* value of X
- Applying $\text{Var}(aX + b) = a^2\text{Var}(X)$ incorrectly to $E[aX + b]$ (forgetting the square)
- Using Binomial when trials are not independent or p is not constant
- Using Poisson without checking that events occur independently at a constant average rate

Summary: Key Formulas from Today

$$p_X(x) = P(X = x), \qquad \sum_x p_X(x) = 1$$

$$F_X(x) = P(X \leq x)$$

$$E[X] = \sum_x x\, p_X(x), \qquad E[aX + b] = aE[X] + b$$

$$\mathrm{Var}(X) = E[X^2] - (E[X])^2, \qquad \mathrm{Var}(aX + b) = a^2\mathrm{Var}(X)$$

Connections Ahead

- Bernoulli $\rightarrow$ building block of Binomial $\rightarrow$ building block of the Naive Bayes likelihood terms from Chapter 2
- Expectation & Variance $\rightarrow$ core machinery for MLE (Week 8) and Confidence Intervals (Week 11)
- Poisson $\rightarrow$ will reappear when modeling rare events in real ML monitoring systems

Preview of Week 4: Continuous Random Variables

- Today: countable outcomes, PMF, sums.
- Next week: uncountable outcomes, PDF, integrals (kept to single-variable calculus only).
- We will meet the Uniform, Exponential, and Gaussian distributions.